Medium Power Phase Control Thyristors (Stud Version), 25 A



TO-48 (TO-208AA)

FEATURES

- Improved glass passivation for high reliability and exceptional stability at high temperature
- High dI/dt and dV/dt capabilities
- Standard package
- Low thermal resistance
- Metric threads version available
- Types up to 1200 V V_{DRM}/V_{RRM}
- Designed and qualified for industrial and consumer level
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

RoHS COMPLIANT

TYPICAL APPLICATIONS

- Medium power switching
- Phase control applications

PRIMARY CHARACTERISTICS	
$I_{T(AV)}$	25 A
V_{DRM}/V_{RRM}	100 V, 200 V, 400 V, 600 V, 800 V, 1000 V 1200 V
V_{TM}	1.70 V
I_{GT}	60 mA
T_J	-65 °C to +125 °C
Package	TO-48 (TO-208AA)
Circuit configuration	Single SCR

MAJOR RATINGS AND CHARACTERISTICS			
PARAMETER	TEST CONDITIONS	VALUES	UNITS
$I_{T(AV)}$		25	A
	T_C	85	°C
$I_{T(RMS)}$		40	A
I_{TSM}	50 Hz	420	A
	60 Hz	440	
I^2t	50 Hz	867	A^2s
	60 Hz	790	
V_{DRM}/V_{RRM}		100 to 1200	V
t_q	Typical	110	µs
T_J		-65 to +125	°C

ELECTRICAL SPECIFICATIONS

VOLTAGE RATINGS				
TYPE NUMBER	VOLTAGE CODE	V_{DRM}/V_{RRM}, MAXIMUM REPETITIVE PEAK AND OFF-STATE VOLTAGE (1) V	V_{RSM}, MAXIMUM NON-REPETITIVE PEAK VOLTAGE (2) V	I_{DRM}/I_{RRM} MAXIMUM AT $T_J = T_J$ MAXIMUM mA
VS-25RIA	10	100	150	20
	20	200	300	10
	40	400	500	
	60	600	700	
	80	800	900	
	100	1000	1100	
	120	1200	1300	

Notes

(1) Units may be broken over non-repetitively in the off-state direction without damage, if dI/dt does not exceed 20 A/µs

(2) For voltage pulses with $t_p \leq 5$ ms

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ABSOLUTE MAXIMUM RATINGS

PARAMETER	SYMBOL	TEST CONDITIONS			VALUES	UNITS
Maximum average on-state current at case temperature	$I_{T(AV)}$	180° sinusoidal conduction			25	A
					85	°C
Maximum RMS on-state current	$I_{T(RMS)}$				40	A
Maximum peak, one-cycle non-repetitive surge current	I_{TSM}	t = 10 ms	No voltage reapplied	Sinusoidal half wave, initial $T_J = T_J$ maximum	420	A
		t = 8.3 ms			440	
		t = 10 ms	100 % V_{RRM} reapplied		350	
		t = 8.3 ms			370	
Maximum I^2t for fusing	I^2t	t = 10 ms	No voltage reapplied		867	A^2s
		t = 8.3 ms			790	
		t = 10 ms	100 % V_{RRM} reapplied		615	
		t = 8.3 ms			560	
Maximum $I^2\sqrt{t}$ for fusing	$I^2\sqrt{t}$	t = 0.1 to 10 ms, no voltage reapplied, $T_J = T_J$ maximum			8670	$A^2\sqrt{s}$
Low level value of threshold voltage	$V_{T(TO)1}$	(16.7 % x π x $I_{T(AV)}$ < I < π x $I_{T(AV)}$), $T_J = T_J$ maximum			0.99	V
High level value of threshold voltage	$V_{T(TO)2}$	(I > π x $I_{T(AV)}$), $T_J = T_J$ maximum			1.40	
Low level value of on-state slope resistance	r_{t1}	(16.7 % x π x $I_{T(AV)}$ < I < π x $I_{T(AV)}$), $T_J = T_J$ maximum			10.1	mΩ
High level value of on-state slope resistance	r_{t2}	(I > π x $I_{T(AV)}$), $T_J = T_J$ maximum			5.7	
Maximum on-state voltage	V_{TM}	I_{pk} = 79 A, T_J = 25 °C			1.70	V
Maximum holding current	I_H	T_J = 25 °C, anode supply 6 V, resistive load			130	mA
Latching current	I_L				200	

SWITCHING

PARAMETER		SYMBOL	TEST CONDITIONS	VALUES	UNITS
Maximum rate of rise of turned-on current	$V_{DRM} \leq 600$ V	dI/dt	$T_J = T_J$ maximum, V_{DM} = Rated V_{DRM} Gate pulse = 20 V, 15 Ω, t_p = 6 µs, t_r = 0.1 µs maximum I_{TM} = (2 x rated dI/dt) A	200	A/µs
	$V_{DRM} \leq 800$ V			180	
	$V_{DRM} \leq 1000$ V			160	
	$V_{DRM} \leq 1600$ V			150	
Typical turn-on time		t_{gt}	T_J = 25 °C, at rated V_{DRM}/V_{RRM}, T_J = 125 °C	0.9	µs
Typical reverse recovery time		t_{rr}	$T_J = T_J$ maximum, $I_{TM} = I_{T(AV)}$, t_p > 200 µs, dI/dt = - 10 A/µs	4	
Typical turn-off time		t_q	$T_J = T_J$ maximum, $I_{TM} = I_{T(AV)}$, t_p > 200 µs, V_R = 100 V, dI/dt = - 10 A/µs, dV/dt = 20 V/µs linear to 67 % V_{DRM}, gate bias 0 V to 100 W	110	

Note

- t_q = 10 µs up to 600 V, t_q = 30 µs up to 1600 V available on special request

BLOCKING

PARAMETER	SYMBOL	TEST CONDITIONS	VALUES	UNITS
Maximum critical rate of rise of off-state voltage	dV/dt	$T_J = T_J$ maximum linear to 100 % rated V_{DRM}	100	V/µs
		$T_J = T_J$ maximum linear to 67 % rated V_{DRM}	300 (1)	

Note

(1) Available with: dV/dt = 1000 V/µs, to complete code add S90 i.e. 25RIA120S90

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TRIGGERING					
PARAMETER	**SYMBOL**	**TEST CONDITIONS**		**VALUES**	**UNITS**
Maximum peak gate power	P_{GM}	$T_J = T_J$ maximum		8.0	W
Maximum average gate power	$P_{G(AV)}$			2.0	
Maximum peak positive gate current	I_{GM}	$T_J = T_J$ maximum		1.5	A
Maximum peak negative gate voltage	$-V_{GM}$	$T_J = T_J$ maximum		10	V
DC gate current required to trigger	I_{GT}	T_J = - 65 °C	Maximum required gate trigger current/voltage are the lowest value which will trigger all units 6 V anode to cathode applied	90	mA
		T_J = 25 °C		60	
		T_J = 125 °C		35	
DC gate voltage required to trigger	V_{GT}	T_J = - 65 °C		3.0	V
		T_J = 25 °C		2.0	
		T_J = 125 °C		1.0	
DC gate current not to trigger	I_{GD}	$T_J = T_J$ maximum, V_{DRM} = Rated value		2.0	mA
DC gate voltage not to trigger	V_{GD}	$T_J = T_J$ maximum, V_{DRM} = Rated value	Maximum gate current/voltage not to trigger is the maximum value which will not trigger any unit with rated V_{DRM} anode to cathode applied	0.2	V

THERMAL AND MECHANICAL SPECIFICATIONS				
PARAMETER	**SYMBOL**	**TEST CONDITIONS**	**VALUES**	**UNITS**
Maximum operating junction and storage temperature range	T_J, T_{Stg}		-65 to +125	°C
Maximum thermal resistance, junction to case	R_{thJC}	DC operation	0.75	K/W
Maximum thermal resistance, case to heat sink	R_{thCS}	Mounting surface, smooth, flat and greased	0.35	
Allowable mounting torque		Non-lubricated threads	$3.4^{+0-10\ \%}$ (30)	N · m (lbf · in)
		Lubricated threads	$2.3^{+0-10\ \%}$ (20)	
Approximate weight			14	g
			0.49	oz.
Case style		See dimensions - link at the end of datasheet	TO-48 (TO-208AA)	

ΔR_{thJC} CONDUCTION				
CONDUCTION ANGLE	**SINUSOIDAL CONDUCTION**	**RECTANGULAR CONDUCTION**	**TEST CONDITIONS**	**UNITS**
180°	0.17	0.13	$T_J = T_J$ maximum	K/W
120°	0.21	0.22		
90°	0.27	0.30		
60°	0.40	0.42		
30°	0.69	0.70		

Note

- The table above shows the increment of thermal resistance R_{thJC} when devices operate at different conduction angles than DC

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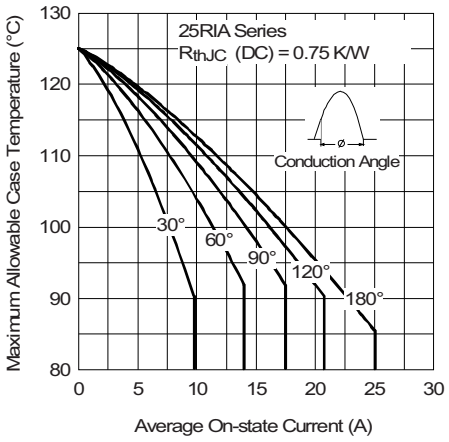


Fig. 1 - Current Ratings Characteristics

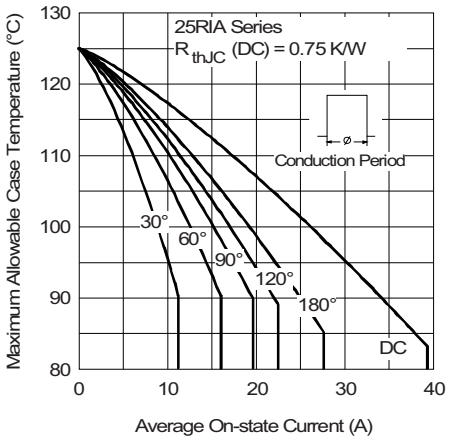


Fig. 1 - Current Ratings Characteristics

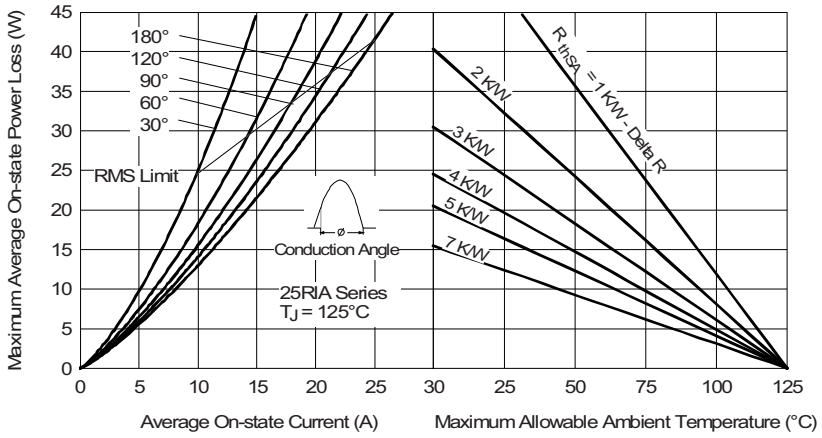


Fig. 2 - On-State Power Loss Characteristics

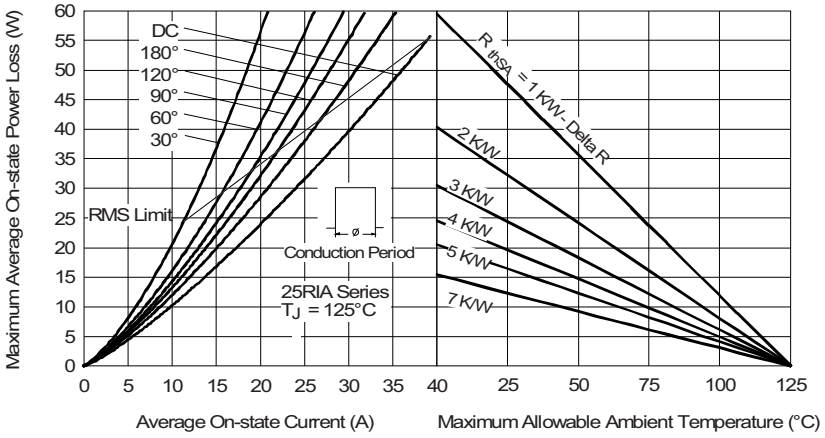


Fig. 3 - On-State Power Loss Characteristics

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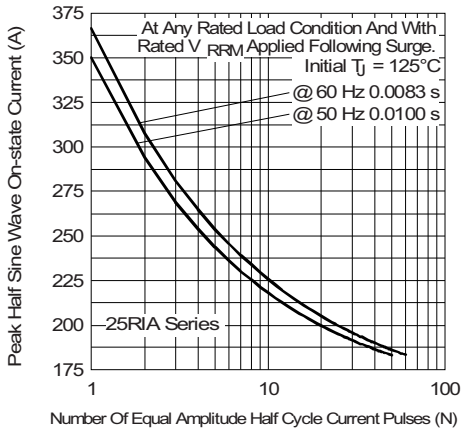


Fig. 4 - Maximum Non-Repetitive Surge Current

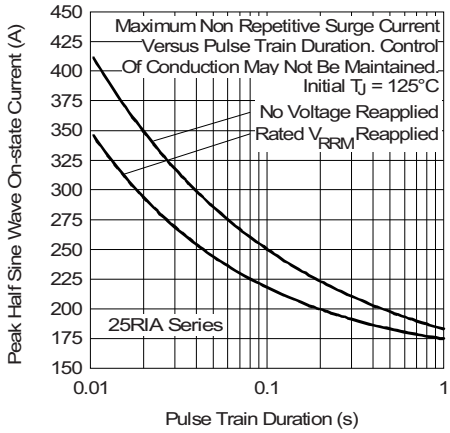


Fig. 5 - Maximum Non-Repetitive Surge Current

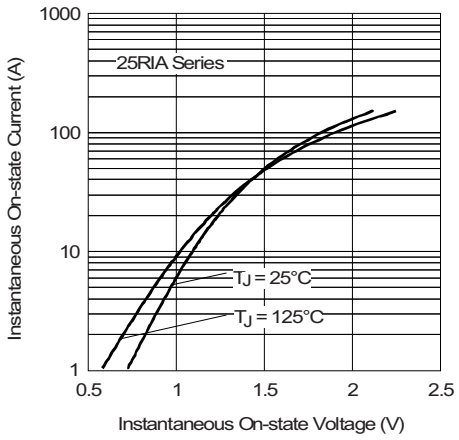


Fig. 6 - Forward Voltage Drop Characteristics

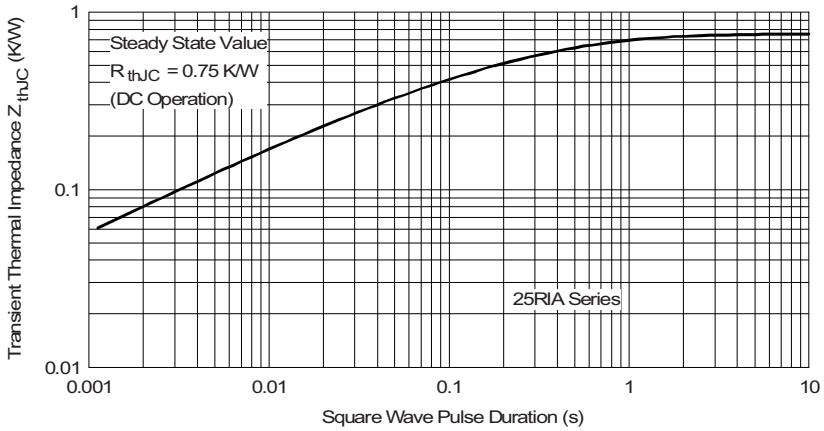


Fig. 7 - Thermal Impedance Z_{thJC} Characteristics

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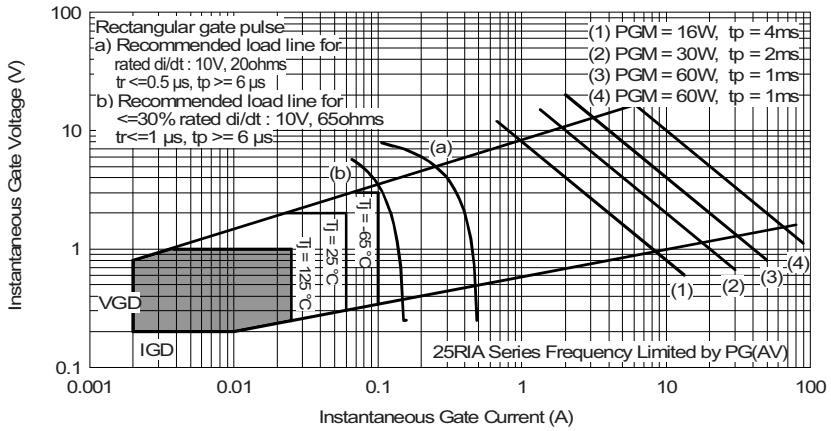


Fig. 8 - Gate Characteristics

ORDERING INFORMATION TABLE

Device code

VS-	25	RIA	120	M	S90
1	2	3	4	5	6

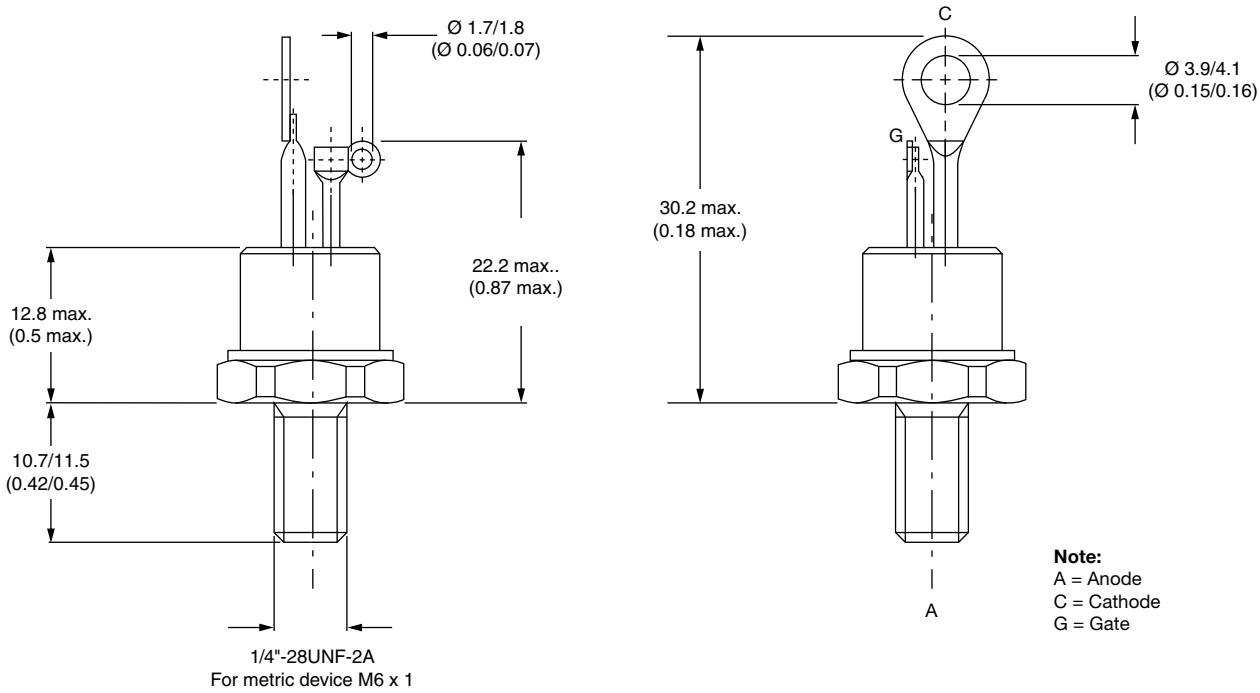
- 1 - Vishay Semiconductors product
- 2 - Current code
- 3 - Essential part number
- 4 - Voltage code x 10 = V_{RRM} (see Voltage Ratings table)
- 5 - None = stud base TO-48 (TO-208AA) 1/4" 28UNF-2A
  M = stud base TO-48 (TO-208AA) M6 x 1
- 6 - Critical dV/dt:
  None = 300 V/µs (standard value)
  S90 = 1000 V/µs (special selection)

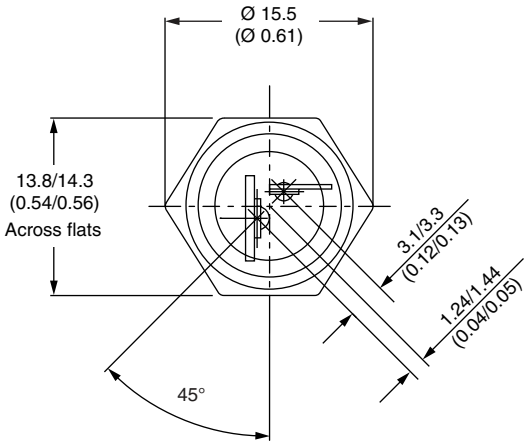
LINKS TO RELATED DOCUMENTS	
Dimensions	www.vishay.com/doc?95333

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TO-208AA (TO-48)

DIMENSIONS in millimeters (inches)





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